

Program: Diploma in Fire Technology and Safety	
Course Code: 6461C	Course Title: Advanced Safety Management
Semester: 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To understand the advanced techniques for safety management in the chemical process industry.
- To learn and apply methods for incident investigation, unavailability analysis, and reliability analysis.
- To gain insight into Layer of Protection Analysis (LOPA) methodology, risk assessment, and safety system design.
- To comprehend security aspects in chemical industries, including site security, cyber security, and countermeasures.
- To learn and implement Behavior-Based Safety (BBS) and understand its importance in safety culture and safety management standards.

Course Pre-requisites:

Topic	Course Code	CourseName	Semester
Principles of Safety Management		Principles of Safety Management	2
Electrical Safety		Electrical Safety	3
Environmental Safety		Environmental Safety	4
Safety in Engineering Industries		Safety in Engineering Industries	4

Course Outcomes

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand advanced safety analysis techniques like Domino Incident Investigation and Reliability Analysis.	14	Understanding
CO2	Apply Layer of Protection Analysis (LOPA) methodology to evaluate and mitigate risks in process systems.	15	Applying
CO3	Analyze and implement security measures for process industries, including physical and cyber security.	14	Analyzing
CO4	Develop and implement Behavior-Based Safety (BBS) programs to promote a safety culture in organizations.	15	Creating
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	3				
CO2	3	2	2				
CO3	3	3	3				
CO4	3	2	3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand advanced safety analysis techniques like Domino Incident Investigation and Reliability Analysis.		
M1.01	Domino incident investigation: technique, logic diagram, input requirements, and output.	3	Understanding
M1.02	Unavailability analysis of protective systems: technique, logic diagram, and example.	4	Understanding

M1.03	Reliability analysis of automatic control systems: PES safety system development, logic diagram, system analysis	3	Applying
M1.04	Introduction to MORT analysis and Sneak analysis.	4	Understanding
CO2	Apply Layer of Protection Analysis (LOPA) methodology to evaluate and mitigate risks in process systems.		
M2.01	Overview of LOPA methodology and relevant standards	4	Understanding
M2.02	Risk tolerance criteria and preparation of LOPA.	4	Applying
M2.03	Developing scenarios, identifying initiating causes and effects.	4	Applying
M2.04	Independent Protection Layers (IPL), SIL assignment, and interpreting LOPA results.	3	Analyzing
	SeriesTestI	1	
Contents: LOPA Methodology: Process of evaluating risk using layers of protection and safety systems. Independent Protection Layers (IPL): Allocation of IPL credit and SIL assignment. Scenario Development: Identifying and analyzing initiating causes and potential effects. Interpreting LOPA Results: Making risk decisions and recommendations.			
CO3	Analyze and implement security measures for process industries, including physical and cyber security.		
M3.01	Assessments and regulatory environment for chemical process industry security.	4	Understanding
M3.02	Methods for assessing security vulnerability and emerging security regulations.	4	Analyzing
M3.03	Threat analysis, security countermeasures, and emergency response strategies.	4	Applying
M3.04	Case study on site security for process industries.	2	Evaluating

Contents:

Security Assessment: Identifying vulnerabilities and emerging regulations.
Threat Analysis and Countermeasures: Developing strategies for preventing intentional releases and theft.
Cyber Security: Securing control systems and addressing risks associated with digital systems.
Site Security: Physical security, mitigation, emergency response, and information security.

CO4	Develop and implement Behavior-Based Safety (BBS) programs to promote a safety culture in organizations.		
M4.01	Fundamentals of BBS: Observation and feedback process.	3	Understanding
M4.02	Critical behavior checklist and outcomes of BBS work	4	Applying
M4.03	Implementation challenges in BBS and Praxis six-cell analysis model.	4	Analyzing
M4.04	Developing a BBS culture: Manager's role and BBS steering committee.	4	Creating
	SeriesTest II	1	

Contents:

BBS Fundamentals: Observations, feedback processes, and checklist for critical behaviors.
Challenges in BBS Implementation: Addressing problems in creating a culture of safety.
Role of Management in BBS: Leading and fostering a safety culture.
BBS Steering Committee: Organizing and maintaining a BBS approach.

Text/Reference:

T/R	BookTitle/Author
R1	Centre for Chemical Process Safety. AIChE, Guidelines for Chemical Process Quantitative Risk Analysis, 2nd Edition, 2000.
R2	Centre for Chemical Process Safety, AIChE, Layer of Protection Analysis – Simplified Process Risk Assessment, 2001.

R3	ACC, Site Security Guidelines for the U.S. Chemical Industry, American Chemistry Council, 2001.
R4	Sam Mannan (Editor), Lee's Loss Prevention in the Process Industries, 4th Edition, Butterworth-Heinemann Ltd., UK, 2012.
R5	Thomas R. Krause, The Behavior-Based Safety Process: Managing Involvement for an Injury-Free Culture, 2nd Edition, John Wiley & Sons, 1996.
R6	Kaila, H.L., Behavior-Based Safety in Organizations – A Practical Guide, IK International, New Delhi, 2010